PROMPT HISTORY & PROJECT DEVELOPMENT LOG

Genetic Algorithm for the Travelling Salesman Problem (GA–TSP)

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About this document

This document records the development of a Genetic Algorithm for the Travelling Salesman Problem, from choosing the algorithm and designing its operators to evaluating results and preparing the notebook, report, and presentation. It includes the original prompts and additional project-specific prompt examples that illustrate more detailed instructions for the same work.

The additional examples are written as part of this expanded project record. The original prompts and the added examples are shown in their respective sections.

1. Project overview

The project implements a Genetic Algorithm (GA) for a Travelling Salesman Problem with 15 Indian cities. Distances are computed from geographic coordinates using the haversine formula. The GA is checked against an exact Held–Karp dynamic-programming solution and compared with random-search and nearest-neighbour baselines. A notebook, LaTeX report, and Beamer presentation form the main deliverables.

Workstream	Question addressed	Output
Topic choice	Which approach is distinctive and defensible in a viva?	GA for TSP chosen instead of the existing A* direction.
Implementation	How can tours be evolved without invalid chromosomes?	Permutation encoding, OX1, inversion mutation, tournament selection, elitism.
Validation	How can route quality be measured objectively?	Held–Karp optimum and baseline comparisons.
Experiments	How do parameters affect performance?	Baseline, mutation-rate, and population-size studies.
Submission	How can another person reproduce the work?	Notebook, figures, report, slides, and project instructions.

Main configuration

• 15 Indian cities; haversine distances with Earth radius 6371 km.

• Population 150; 300 generations; tournament selection k = 5.

• OX1 crossover probability 0.9; inversion mutation probability 0.2; elitism preserves the best 2.

• Held–Karp exact optimum reported as 7478.9 km; multiple seeds used for evaluation.

2. Topic selection and notebook development

Prompt 1 — Initial request

> “go through the handout and help me with the 10 makrer component of algorithm implementation and viva i am planning you give me first a jupyter nb file or something while would be hands down implementation type stuff then we compile a presentable latex doc with screenshot and all of the methodology”

Outcome: The project began as a two-stage workflow: first build an implementation-focused notebook, then prepare a LaTeX report with screenshots and methodology. An existing A* grid-pathfinding direction was considered before the topic was settled.

Prompt 2 — Rejecting A*

> “i dont think so a* would be noble”

Outcome: The intent was that A* did not feel novel enough. The direction changed to GA for TSP, which supports route visualisation and objective validation on a manageable instance.

Additional project prompt — topic evaluation

> “Review the assignment handout first. Compare possible topics from the allowed AI approaches by implementation effort, visual clarity, novelty, ease of viva explanation, and ability to validate results. Recommend a direction and explain the trade-offs before editing files. I want the project to demonstrate algorithmic implementation rather than simply calling a library.”

Additional project prompt — define the problem

> “Use a Genetic Algorithm for the Travelling Salesman Problem over 15 Indian cities. Use latitude and longitude coordinates, compute pairwise distances with the haversine formula, and make the distance matrix symmetric with a zero diagonal. Explain why a tour should be encoded as a permutation and why this is more appropriate than a binary chromosome for this problem.”

3. Implementation and validation prompts

Prompt — data and distance matrix

“Create a table for Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad, Pune, Jaipur, Lucknow, Bhopal, Patna, Chandigarh, Guwahati, and Kochi. Store coordinates clearly. Implement haversine distance with R = 6371 km, create the 15 × 15 matrix, and add checks for symmetry, zero diagonal, and non-negative entries. Print a compact summary so data errors can be caught before running the GA.”

Prompt — fitness and tour length

“Implement tour_length so the final city connects back to the starting city. Use fitness = 1 / tour length for selection, and document the relationship between minimising distance and maximising fitness. Add tests using a small hand-constructed tour and ensure the function does not accidentally omit the return leg.”

Prompt — evolution loop and reproducibility

“Write genetic_algorithm using population size 150 and 300 generations. Fix random seeds, preserve the best two tours, and track best-so-far and population-average distance at every generation. Return the best tour, its distance, convergence history, and runtime. Separate algorithm logic from plotting so the core algorithm can be tested independently.”

Prompt — validity checks

“Add assertions that every chromosome contains each city index exactly once, every tour length is finite and positive, and each genetic operator returns a valid permutation. Include a short smoke test before the full run. If a test fails, stop with a clear error rather than continuing to generate misleading plots.”

Prompt — visualisations

“Plot the best route using longitude and latitude, connect cities in visit order, label the start city, and keep the legend and axes readable. Plot convergence separately, showing best-so-far and population-average distance. Save figures to figures/ with descriptive filenames so they can be inserted into the report.”

Prompt — exact optimum with Held–Karp

“Implement Held–Karp dynamic programming to find the exact optimum for the 15-city instance. Explain the DP state, recurrence, base case, and reconstruction of the route. Compare the GA route to the optimum using absolute distance and percentage gap. State clearly that GA is stochastic and is not guaranteed to find the optimum in every run.”

Prompt — explain crossover for a viva

“Add a worked OX1 example using P1 = [1,2,3,4,5,6,7,8], P2 = [3,7,5,1,6,8,2,4], and cut positions 2 through 4. Show the slice copied from one parent and the order in which remaining genes are taken from the other parent, skipping duplicates. Add an inversion mutation example and explain why both operators preserve a permutation.”

> **Additional project prompt - design operators**
>
> "Implement tournament selection with k = 5, Order Crossover (OX1) with probability 0.9, inversion mutation with probability 0.2, and elitism of two individuals. Keep each operator in a separate function, comment the logic, and test that crossover and mutation always preserve valid permutations. Do not use a GA library."

4. Experimentation and result analysis

Prompt — baseline comparison

> “Compare GA over ten seeds with nearest neighbour and random search using the same nominal evaluation budget of 150 × 300; average random search over three seeds. Report best and mean tour length, gap from the exact optimum, and how many GA runs reach the optimum. Save the result table to CSV and generate a clear comparison plot.”

Prompt — mutation-rate experiment

> “Test mutation rates 0, 0.05, 0.2, 0.5, and 0.9 with eight seeds per rate and 200 generations. Keep other parameters fixed, store all final distances, and make a box plot with the exact optimum as a reference line. Discuss diversity or stagnation only where the results support the interpretation.”

Prompt — population-size experiment

> “Compare population sizes 10, 30, 100, and 300 with an equal budget of 45,000 evaluations and eight seeds for each size. Record each run, save a CSV, and show distributions in a box plot. Do not compare settings using only a single favourable seed.”

Prompt — honest result reporting

> “Generate report text and tables from computed values rather than hard-coding claims. Preserve weaker runs as evidence of variability; report the seed-42 result even if it is worse than the best run. If the GA does not reach the exact optimum, show the gap and explain possible local-optimum behaviour without hiding the result.”

Prompt — complexity and limitations

> “Explain GA time complexity as O(G·P·n) and space as O(P·n). Explain Held–Karp time as O(n²·2■) and space as O(n·2■). Include the lack of an optimality guarantee, dependence on seeds and parameter settings, and the possibility of premature convergence. Keep the explanation precise enough for an oral defence.”

Recorded result checkpoints

Method	Length (km)	Gap
Random search (mean of 3 seeds)	9930.4	32.8%
Nearest neighbour	9169.4	22.6%
GA (mean of 10 runs)	7526.9	0.6%
GA (best of 10 runs)	7478.9	0%
Held–Karp optimum	7478.9	0%

The supplied project history states that the GA reached the optimum in 5 of 10 runs. These figures are included as recorded checkpoints.

5. Report and presentation development

Prompt — notebook polish and report

> “get rid of the viva questions section keep this nb presentable as well and start with the compiling of the latex file use the figures you picked i have added the college logo as well my name is Shashwat Shikhar Dwivedi roll no 2427030605 Submitted to Dr.Neha Chaudhary”

Outcome: The viva-question section was removed, the notebook was polished, and the report was prepared with the provided identity details, selected figures, and college logo. The supplied history says the notebook ran through 23 cells without errors. The report included methodology, crossover example, pseudocode, Held–Karp validation, results, complexity, limitations, and conclusion.

Additional project prompt — report quality

> “Prepare an A4, 11-point LaTeX report with abstract, contents, introduction, methodology, implementation excerpts, results, complexity, limitations, and conclusion. Place figures near their interpretation, include units and percentage gaps, and make sure every table and claim traces back to notebook outputs. Use the college logo and my supplied details on the title page.”

Original presentation prompt — Beamer template

> A Beamer help article was pasted, followed by: “use this latex beamer template and give me presentable ppt or pdf compiled from it which i can turn in”

Outcome: The first deck focused on Beamer itself because the intended subject was not stated. This was corrected in the next prompt.

Original presentation prompt — correct the subject

> “about the topic itslef like the project we have made ga tsp one”

Outcome: The presentation was rebuilt around the GA–TSP project as a 17-slide Beamer deck, covering TSP, GA design, crossover, exact optimum versus GA, convergence, baselines, experiments, complexity, and limitations.

Additional project prompt — slide consistency

> “Create 15–17 slides with the Madrid theme and beaver colour theme. Keep each slide focused, use figures from ../figures/, and make every number match the notebook. Include the worked crossover example, baseline table, parameter experiments, limitations, and a clear conclusion. Put my full name and roll number on the title slide and full name in the footer.”

Original presentation prompt — full name correction

> “change the name to Shashwat Shikhar Dwivedi”

Outcome: The shortened footer name was replaced with the full name. The supplied history notes that visual inspection of the recompiled slides was still important.

6. Project documentation and file structure

Prompt — prompt history for the instructor

> “i am supposed to submitt prompt history as well to maam she was saying she should be able to replicate the project file so please help me here”

Outcome: This led to preparing the prompt-history document and a separate consolidated replication prompt so the instructor could follow the project workflow and reproduce the outputs.

Prompt — improve the prompt document

> “no like what i have done is like very basic stuff in prompting but like she would grade me on the basis of that so give me better promt to submitt to maam”

Outcome: The prompt record was expanded with additional project-specific technical instructions and a more detailed account of the workflow.

Project file map

• **Notebook:** GA_TSP_Algorithmic_Implementation.ipynb — run all cells to regenerate calculations, figures, and CSV outputs.

• **Figures:** figures/ — plots and experiment tables used by the report and presentation.

• **Logo:** college_logo.jpg — report title-page asset.

• **Report:** report.tex → report.pdf; compile from the project root using tectonic report.tex or pdflatex report.tex twice.

• **Slides:** beamer/presentation.tex → presentation.pdf; compile from beamer using tectonic presentation.tex; figures are expected at ../figures/.

• **Replication prompt:** REPLICATION_PROMPT.md/PDF — project requirements, experiments, and expected-result checkpoints.